

Attention

A small English-to-French transformer, built the way Vaswani et al. described it in 2017: multi-head attention, sinusoidal positions, and a position-wise feed-forward network.

1,008,000 parameters · 2 layers · d_model 128 · 4 heads · 387 training sentences · validation token accuracy 67.8%

TranslateGo.Run!Hello.Thank you.How are you?I love you.Wait!Help!Good morning.He is a doctor.This is a book.

French

Je t'aime !

Read as i love you .

Steps

These figures are measured on the sentence you just translated.

1. Token ids

Token ids from the English vocabulary: i → 10, love → 200, you → 17, . → 4.

token	id	in vocabulary
i	10	yes
love	200	yes
you	17	yes
.	4	yes

2. Embedding plus position

Each id is a 128-dimensional vector multiplied by $\sqrt{128} = 11.314$. A sinusoid of the same width is then added. The table shows those lengths and the first four coordinates. Under it, the curves and the grid use the same dimensions.

If the love embedding is

[-0.0864, 0.0873, -0.1197, ...]

then after scaling, the first coordinates are

[-0.0864×11.314 , 0.0873×11.314 , -0.1197×11.314 , ...]

which is about

[-0.977 , 0.987 , -1.354 , ...]

The table value $\|\text{word} \times \sqrt{d}\| = 10.798$ is the length of all 128 coordinates:

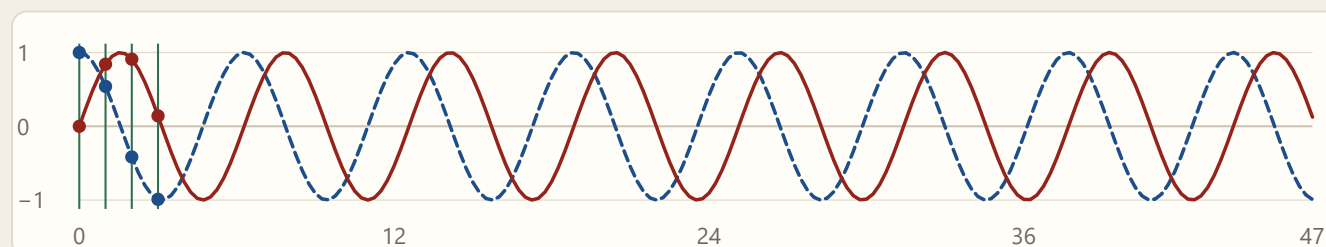
$$\sqrt{(x_1^2 + x_2^2 + \dots + x_{128}^2)} = \sqrt{116.603} = 10.798$$

token	$\ \text{word} \times \sqrt{d}\ $	$\ \text{position}\ $	$\ \text{sum}\ $	position dims 0–3	sum dims 0–3
i	10.059	8.000	11.646	0.000, 1.000, 0.000, 1.000	-0.407, -0.387, -0.698, 0.3
love	10.798	8.000	11.442	0.841, 0.540, 0.762, 0.648	-0.136, 1.528, -0.593, -0.5
you	9.793	8.000	12.216	0.909, -0.416, 0.987, -0.160	1.222, -0.955, -0.151, -1.19
.	8.677	8.000	10.054	0.141, -0.990, 0.517, -0.856	0.896, -1.327, 0.758, -0.01

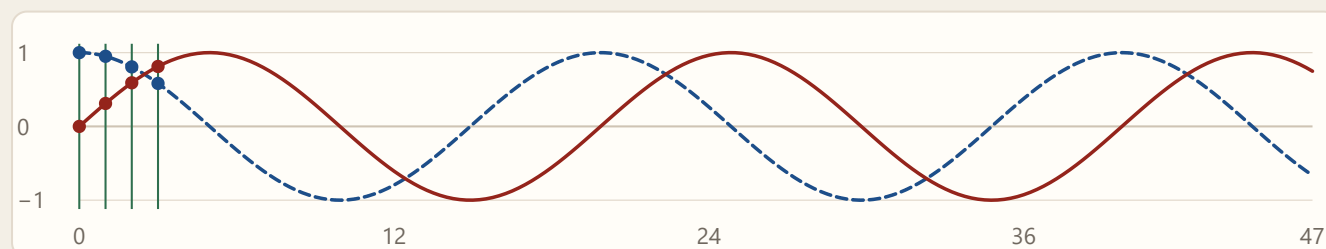
— sine, even dimension - - - cosine, odd dimension — this sentence

$\text{PE}(\text{position}, 2i) = \sin(\text{position} / 10000^{(2i/128)})$, and dimension $2i+1$ is the cosine. Dimension 0 repeats every 6.28 positions; dimension 64 repeats every 628.32. Ticks mark i at position 0, love at position 1, you at position 2, . at position 3.

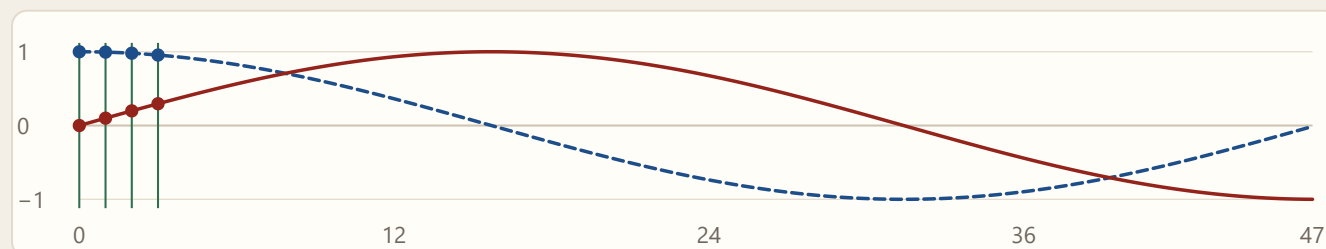
Dimensions 0 and 1. Repeats every 6.28 positions.



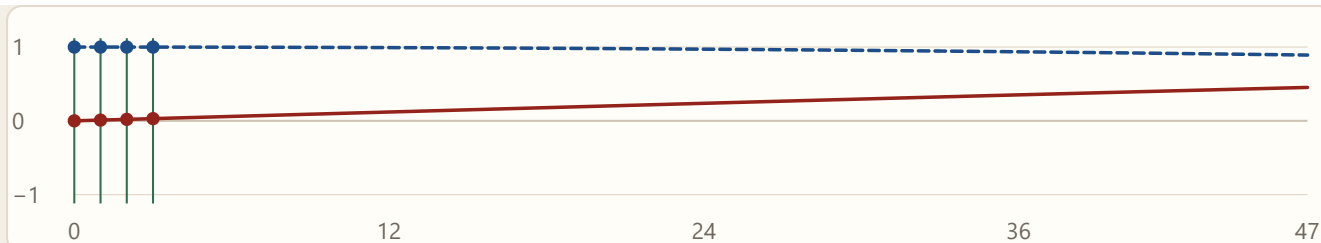
Dimensions 16 and 17. Repeats every 19.87 positions.



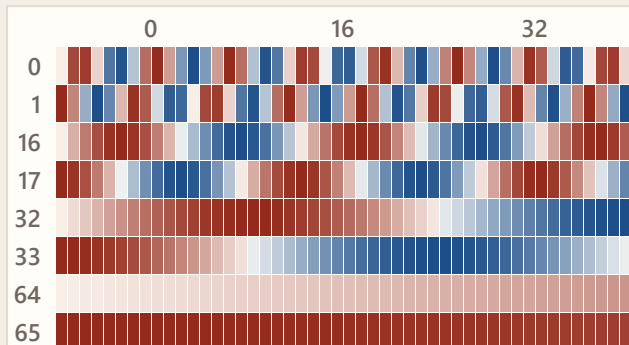
Dimensions 32 and 33. Repeats every 62.83 positions.



Dimensions 64 and 65. Repeats every 628.32 positions.



Dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the curves, at positions 0–47. Even rows are sines and odd rows are cosines.



Rows are dimensions 0, 1, 16, 17, 32, 33, 64, 65. Columns are positions 0–47. Even rows are sine and odd rows are cosine. Color is scaled to the largest absolute value here, 1.000. Hover a cell for the value. Rust is positive and blue is negative.

Scaled word vectors at those dimensions

	0	1	16	17	32	33	64	65
i	-0.41	-1.39	-1.04	-1.35	0.43	-1.13	0.36	0.92
love	-0.98	0.99	-0.28	-1.66	-0.23	-0.61	-0.64	-0.72
you	0.31	-0.54	1.21	-2.21	-1.13	0.67	0.54	-1.74
.	0.76	-0.34	0.99	-1.22	0.03	0.13	0.90	-1.02

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.208. Hover a cell for the value. Rust is positive and blue is negative.

Positional vectors at those dimensions

	0	1	16	17	32	33	64	65
i	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00
love	0.84	0.54	0.31	0.95	0.10	0.99	0.01	1.00
you	0.91	-0.42	0.59	0.81	0.20	0.98	0.02	1.00
.	0.14	-0.99	0.81	0.58	0.30	0.95	0.03	1.00

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 1.000. Hover a cell for the value. Rust is positive and blue is negative.

Sum that enters the first encoder layer, same dimensions

	0	1	16	17	32	33	64	65
i	-0.41	-0.39	-1.04	-0.35	0.43	-0.13	0.36	1.92
love	-0.14	1.53	0.03	-0.71	-0.13	0.38	-0.63	0.28

you	1.22	-0.95	1.80	-1.40	-0.93	1.65	0.56	-0.74
.	0.90	-1.33	1.80	-0.64	0.33	1.09	0.93	-0.02

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 1.916. Hover a cell for the value. Rust is positive and blue is negative.

3. Encoder layers

Last of 2 encoder layers, mean of 4 heads. i attends most to i (0.5956) love attends most to love (0.7990) you attends most to love (0.9655) . attends most to love (0.9957). Both layers follow, with every head, the ReLU hidden state, and the residual stream.

Encoder layer 1

At i, 95 of 512 ReLU units are on. Residual length 11.422 after attention and 11.448 after the feed-forward block.

Self-attention. Rows and columns are the English tokens.

all heads, mean ▾

	i	love	you	.
i	0.329	0.078	0.147	0.447
love	0.287	0.261	0.451	0.001
you	0.258	0.357	0.359	0.026
.	0.009	0.891	0.085	0.015

Residual stream after attention

	0	1	16	17	32	33	64	65
i	0.17	-0.33	-2.44	-1.29	-0.07	-0.95	0.65	0.99
love	-0.23	1.54	-1.43	-1.18	-0.80	1.24	-0.50	0.03
you	0.67	-0.12	-0.09	-2.12	-1.36	2.23	0.37	-1.07
.	-0.07	-1.05	0.86	-2.11	0.37	2.23	1.10	-1.14

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.443. Hover a cell for the value. Rust is positive and blue is negative.

Feed-forward ReLU, i 95/512, love 89/512, you 88/512, . 101/512 units on

	0	1	16	17	32	33	64	65
i	0.00	0.00	0.00	1.24	2.44	0.00	0.00	0.00
love	0.00	0.00	0.00	1.33	1.52	0.00	0.00	0.00
you	0.00	0.00	0.00	1.38	0.79	0.00	0.00	0.00
.	0.00	0.00	1.36	0.95	0.75	0.00	0.00	0.00

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.441. Hover a cell for the value. Rust is positive and blue is negative.

Residual stream after the feed-forward block

	0	1	16	17	32	33	64	65
i	-0.15	-0.40	-1.28	-1.74	0.36	-1.01	2.23	1.09
love	-0.28	0.99	-0.99	-0.85	-0.73	1.87	-0.32	-0.46
you	0.49	0.34	-0.22	-2.02	-1.14	2.25	0.03	-1.70
.	-0.38	-1.16	0.98	-1.28	0.93	2.09	-0.09	-1.86

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.250. Hover a cell for the value. Rust is positive and blue is negative.

Encoder layer 2

At i, 233 of 512 ReLU units are on. Residual length 11.374 after attention and 10.477 after the feed-forward block.

Self-attention. Rows and columns are the English tokens.

all heads, mean ▾

	i	love	you	.
i	0.596	0.253	0.100	0.051
love	0.074	0.799	0.051	0.075
you	0.023	0.966	0.010	0.002
.	0.002	0.996	0.002	0.000

Residual stream after attention

	0	1	16	17	32	33	64	65
i	-0.30	-0.87	-1.11	-1.51	-0.64	-1.18	1.98	0.28
love	-0.33	1.05	0.29	-1.63	-1.12	1.35	-0.36	-0.74
you	-0.30	0.67	0.53	-2.28	-1.39	1.46	-0.19	-1.41
.	-0.81	-0.10	1.21	-1.91	-0.21	1.42	-0.26	-1.53

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.277. Hover a cell for the value. Rust is positive and blue is negative.

Feed-forward ReLU, i 233/512, love 189/512, you 182/512, . 150/512 units on

	0	1	16	17	32	33	64	65
i	0.00	0.73	0.72	0.97	0.00	2.17	0.00	0.86
love	0.00	0.85	0.00	0.00	0.00	0.29	0.67	0.00

you	0.00	0.49	0.00	0.00	0.00	0.33	0.48	0.00
.	0.00	0.62	0.00	0.15	0.00	0.00	0.00	0.00

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.173. Hover a cell for the value. Rust is positive and blue is negative.

Residual stream after the feed-forward block

	0	1	16	17	32	33	64	65
i	-1.39	0.09	-0.93	-1.11	-0.26	-2.30	0.99	0.90
love	-1.24	1.22	-0.02	-1.77	-0.57	-1.11	-0.18	-0.26
you	-1.30	1.10	-0.33	-2.14	-0.63	-1.01	-0.11	-0.49
.	-1.26	0.56	0.27	-1.84	-0.08	-0.89	-0.12	-0.51

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.300. Hover a cell for the value. Rust is positive and blue is negative.

4. Decoder layers on the chosen sentence

One forward pass of the prefix the beam kept, including the step that writes <eos>. Rows of the residual and ReLU maps are the French token that position writes.

Decoder layer 1

While writing je, 110 of 512 ReLU units are on. Residual length 12.31 after masked self-attention, 11.484 after cross-attention, and 11.524 after the feed-forward block.

Masked self-attention. The axes are the tokens already fed to the decoder, starting with <bos>.

all heads, mean ▾

	<bos>	je	t	'	aime	!
<bos>	1.000	0.000	0.000	0.000	0.000	0.000
je	0.500	0.500	0.000	0.000	0.000	0.000
t	0.247	0.253	0.500	0.000	0.000	0.000
'	0.250	0.000	0.250	0.499	0.000	0.000
aime	0.250	0.000	0.251	0.234	0.265	0.000
!	0.000	0.000	0.000	0.002	0.000	0.998

Cross-attention. Each row is the French token being written; columns are the English tokens.

all heads, mean ▾

i	love	you	.
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je	0.363	0.097	0.151	0.389
t	0.215	0.112	0.200	0.473
'	0.163	0.225	0.256	0.356
aime	0.105	0.415	0.201	0.279
!	0.233	0.547	0.139	0.082
<eos>	0.375	0.159	0.188	0.278

Residual stream after masked self-attention

	0	1	16	17	32	33	64	65
je	-0.06	1.72	-1.27	-0.70	-0.98	0.15	0.76	-0.13
t	-0.47	0.80	-1.19	0.61	0.45	-0.88	0.88	0.47
'	-0.72	-0.57	0.28	0.57	0.47	0.37	-0.69	-0.23
aime	0.79	-1.44	1.02	-0.67	0.39	0.54	-0.35	-0.06
!	-0.56	-2.71	1.02	-2.04	0.88	0.18	-0.50	-0.26
<eos>	2.02	-0.64	0.93	0.43	-1.57	0.74	0.82	0.76

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.712. Hover a cell for the value. Rust is positive and blue is negative.

Residual stream after cross-attention

	0	1	16	17	32	33	64	65
je	0.36	0.74	-1.84	0.01	-0.67	-0.08	0.47	-0.47
t	0.08	-0.08	-1.82	0.83	0.24	-0.97	0.65	-0.03
'	-0.00	-1.30	-0.67	0.68	0.23	-0.06	-0.62	-0.52
aime	1.17	-2.16	-0.21	-0.19	0.12	0.13	-0.13	-0.41
!	0.28	-2.96	-0.05	-1.41	0.60	0.00	-0.41	-0.37
<eos>	2.17	-1.29	-0.24	0.81	-1.16	0.50	0.65	0.18

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.961. Hover a cell for the value. Rust is positive and blue is negative.

Feed-forward ReLU, je 110/512, t 126/512, ' 100/512, aime 137/512, ! 199/512, <eos> 236/512 units on

	0	1	16	17	32	33	64	65
je	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
t	0.00	0.00	0.00	0.00	2.40	0.00	0.00	0.00
'	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
aime	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.46
!	0.93	0.00	0.00	0.00	0.00	0.00	0.00	0.00

<eos>	0.00	0.00	0.00	1.91	0.00	0.00	0.00	0.00
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Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.405. Hover a cell for the value. Rust is positive and blue is negative.

Residual stream after the feed-forward block

	0	1	16	17	32	33	64	65
je	-2.08	0.53	-1.37	-0.86	0.27	-0.53	1.02	1.39
t	-1.55	-0.18	-1.35	0.26	-0.32	-0.48	0.79	1.22
'	-0.14	-1.08	0.49	-0.16	1.14	0.20	-0.71	1.60
aime	1.48	-1.47	-0.65	-0.25	-0.58	-1.36	-1.13	-1.02
!	1.01	-2.27	1.24	0.12	0.16	0.09	0.15	-0.07
<eos>	1.86	0.23	0.09	-0.51	-1.36	1.03	1.53	1.01

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.271. Hover a cell for the value. Rust is positive and blue is negative.

Decoder layer 2

While writing je, 126 of 512 ReLU units are on. Residual length 11.468 after masked self-attention, 11.305 after cross-attention, and 13.163 after the feed-forward block.

Masked self-attention. The axes are the tokens already fed to the decoder, starting with <bos>.

all heads, mean ▾

	<bos>	je	t	'	aime	!
<bos>	1.000	0.000	0.000	0.000	0.000	0.000
je	0.000	1.000	0.000	0.000	0.000	0.000
t	0.000	0.000	1.000	0.000	0.000	0.000
'	0.000	0.000	0.001	0.999	0.000	0.000
aime	0.000	0.000	0.000	0.000	1.000	0.000
!	0.000	0.000	0.000	0.000	0.000	1.000

Cross-attention. Each row is the French token being written; columns are the English tokens.

all heads, mean ▾

	i	love	you	.
je	0.990	0.004	0.005	0.001
t	0.009	0.327	0.399	0.266

'	0.003	0.346	0.255	0.396
aime	0.285	0.212	0.176	0.327
!	0.013	0.816	0.112	0.060
<eos>	0.241	0.084	0.159	0.516

Residual stream after masked self-attention

	0	1	16	17	32	33	64	65
je	-2.61	0.79	-1.42	-1.37	-0.30	-1.68	0.50	1.64
t	-1.59	0.16	-1.28	-0.20	-0.72	-1.10	0.37	1.80
'	-0.25	-0.91	1.98	-0.10	1.60	0.71	-1.27	1.54
aime	0.69	-1.24	-0.92	-0.58	-0.61	-2.18	-0.81	0.42
!	0.93	-0.94	0.27	0.78	-0.47	0.49	0.98	-0.29
<eos>	2.21	0.07	-0.13	0.15	-0.65	0.54	1.43	0.12

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.614. Hover a cell for the value. Rust is positive and blue is negative.

Residual stream after cross-attention

	0	1	16	17	32	33	64	65
je	-3.25	0.11	-0.89	-0.92	0.60	-2.30	0.53	0.97
t	-3.13	-0.41	-1.40	0.98	0.22	-1.71	0.43	1.48
'	-2.33	-1.17	0.81	1.10	1.75	-0.59	-0.67	1.34
aime	-1.29	-1.26	-0.86	0.72	0.43	-2.67	-0.52	0.69
!	-1.42	-1.03	-0.42	1.50	0.35	-0.54	0.85	0.27
<eos>	-0.43	-0.28	-0.36	0.99	0.28	-0.97	1.14	0.34

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 3.248. Hover a cell for the value. Rust is positive and blue is negative.

Feed-forward ReLU, je 126/512, t 134/512, ' 167/512, aime 150/512, ! 211/512, <eos> 285/512 units on

	0	1	16	17	32	33	64	65
je	0.00	1.28	0.19	0.00	0.81	0.00	0.00	1.29
t	0.00	1.36	0.18	0.00	0.00	0.00	0.00	0.48
'	0.00	0.57	0.00	0.00	0.00	0.00	0.00	2.21
aime	0.00	1.02	1.23	0.00	0.17	0.00	0.00	0.29
!	0.00	0.41	0.00	0.20	0.66	0.00	0.00	0.26
<eos>	0.00	0.49	0.97	0.00	0.00	0.31	0.86	1.53

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.210. Hover a cell for the value. Rust is positive and blue is

negative.

Residual stream after the feed-forward block

	0	1	16	17	32	33	64	65
je	-1.25	1.15	-1.00	-0.23	1.75	-2.75	0.78	0.67
t	-1.91	0.16	-0.03	0.11	-0.33	-1.18	-0.39	0.21
,	0.27	-0.71	3.00	-0.63	0.40	0.79	-1.03	0.76
aime	0.68	-1.35	0.44	-0.67	-0.15	-1.29	-0.13	-0.72
!	0.27	-1.14	1.27	-0.99	-1.60	0.63	1.57	0.68
<eos>	2.35	0.64	1.55	-0.00	0.46	2.00	0.20	-1.57

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65, the same samples as the sinusoid curves. Color is scaled to the largest absolute value here, 2.995. Hover a cell for the value. Rust is positive and blue is negative.

Target word vectors scaled by $\sqrt{d}$, same dimensions as the sinusoids

	0	1	16	17	32	33	64	65
<bos>	-0.01	0.25	-0.35	-0.15	0.14	0.31	-0.04	-1.57
je	-1.72	1.52	-0.79	0.77	1.64	-1.40	0.72	0.11
t	-1.47	0.90	-0.35	0.68	-0.63	0.12	-1.13	-0.88
,	0.07	0.26	0.97	-0.16	-0.25	0.96	-0.81	0.72
aime	0.59	-1.40	-1.13	-0.68	0.47	-1.59	-0.96	-1.16
!	0.36	-0.48	1.17	0.16	-0.99	0.84	1.12	0.27

Columns are dimensions 0, 1, 16, 17, 32, 33, 64, 65. Color is scaled to the largest absolute value here, 1.722. Hover a cell for the value. Rust is positive and blue is negative.

5.1 Decoder writes je

The decoder has read <bos>. je gets probability 0.9258 (92.6%), from log probability -0.0771. Its beam score is $-0.0771 / 1.0000 = -0.0771$, using length penalty $((5+1)/6)^{0.6}$.

candidate	probability	log probability
je	0.9258	-0.0771
demande	0.0018	-6.2962
t	0.0014	-6.5539
aha	0.0014	-6.5669
quelle	0.0009	-7.0570

Beam kept after this choice. The highlighted row is the one that becomes the translation. Score = log probability / length penalty.

hypothesis	log P	length penalty	score
je	-0.0771	1.0000	-0.0771

demande	-6.2962	1.0000	-6.2962
t	-6.5539	1.0000	-6.5539
aha	-6.5669	1.0000	-6.5669

5.2 Decoder writes t

The decoder has read <bos> je. t gets probability 0.9537 (95.4%), from log probability -0.0474. Its beam score is $-0.1245 / 1.0969 = -0.1135$, using length penalty $((5+2)/6)^{0.6}$.

candidate	probability	log probability
t	0.9537	-0.0474
ceci	0.0017	-6.3948
compatis	0.0007	-7.2814
tomber	0.0007	-7.2847
imbécile	0.0006	-7.3686

Beam kept after this choice. The highlighted row is the one that becomes the translation. Score = log probability / length penalty.

hypothesis	log P	length penalty	score
je t	-0.1245	1.0969	-0.1135
je ceci	-6.4719	1.0969	-5.9002
t '	-6.6116	1.0969	-6.0275
aha !	-6.6463	1.0969	-6.0591

5.3 Decoder writes '

The decoder has read <bos> je t. ' gets probability 0.9521 (95.2%), from log probability -0.0491. Its beam score is $-0.1736 / 1.1884 = -0.1461$, using length penalty $((5+3)/6)^{0.6}$.

candidate	probability	log probability
'	0.9521	-0.0491
aime	0.0008	-7.0775
prudemment	0.0008	-7.1683
t	0.0008	-7.1691
reviens	0.0006	-7.4527

Beam kept after this choice. The highlighted row is the one that becomes the translation. Score = log probability / length penalty.

hypothesis	log P	length penalty	score
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je t '	-0.1736	1.1884	-0.1461
t ' aime	-6.6684	1.1884	-5.6113
aha ! <eos>	-6.7138	1.1884	-5.6495
je t aime	-7.2020	1.1884	-6.0602

5.4 Decoder writes aime

The decoder has read <bos> je t '. aime gets probability 0.9408 (94.1%), from log probability -0.0610. Its beam score is $-0.2346 / 1.2754 = -0.1840$, using length penalty $((5+4)/6)^{0.6}$.

candidate	probability	log probability
aime	0.9408	-0.0610
!	0.0024	-6.0404
'	0.0009	-7.0196
je	0.0006	-7.3447
suggestions	0.0006	-7.3471

Beam kept after this choice. The highlighted row is the one that becomes the translation. Score = log probability / length penalty.

hypothesis	log P	length penalty	score
je t ' aime	-0.2346	1.2754	-0.1840
je t ' !	-6.2140	1.2754	-4.8721
t ' aime !	-6.7366	1.2754	-5.2818
je t ' '	-7.1932	1.2754	-5.6398

5.5 Decoder writes !

The decoder has read <bos> je t ' aime. ! gets probability 0.9338 (93.4%), from log probability -0.0685. Its beam score is $-0.3031 / 1.3587 = -0.2231$, using length penalty $((5+5)/6)^{0.6}$.

candidate	probability	log probability
!	0.9338	-0.0685
aime	0.0008	-7.0844
?	0.0005	-7.6584
t	0.0005	-7.6852
je	0.0004	-7.7381

Beam kept after this choice. The highlighted row is the one that becomes the translation. Score = log probability / length penalty.

hypothesis	log P	length penalty	score
je t ' aime !	-0.3031	1.3587	-0.2231
je t ' ! <eos>	-6.2814	1.3587	-4.6232
t ' aime ! <eos>	-6.8039	1.3587	-5.0078
je t ' ' aime	-7.2544	1.3587	-5.3394

5.6 Decoder writes <eos>

The decoder has read <bos> je t ' aime !. <eos> gets probability 0.9342 (93.4%), from log probability -0.0681. Its beam score is $-0.3712 / 1.4386 = -0.2580$, using length penalty $((5+6)/6)^{0.6}$.

candidate	probability	log probability
<eos>	0.9342	-0.0681
aime	0.0003	-7.9831
t	0.0003	-8.0663
'	0.0003	-8.1250
?	0.0003	-8.1334

Beam kept after this choice. The highlighted row is the one that becomes the translation. Score = log probability / length penalty.

hypothesis	log P	length penalty	score
je t ' aime ! <eos>	-0.3712	1.4386	-0.2580
je t ' ' aime !	-7.3246	1.4386	-5.0914
je t ' aime ! aime	-8.2862	1.4386	-5.7598
je t ' aime ! t	-8.3693	1.4386	-5.8176

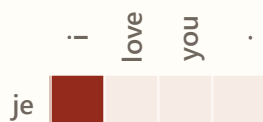
6. Join the tokens

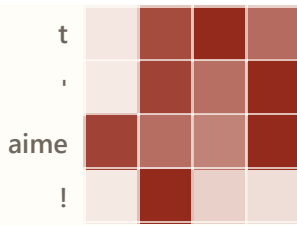
je t ' aime ! → Je t'aime !

Cross-attention

Each row is a French token. The color is how strongly the last decoder layer looked at each English token while writing it. Color is scaled within the row; the tooltip is the real weight.

Head all heads, mean ▼





$\text{softmax}(Q K^T / \sqrt{d_k}) V$

Training sentences are short pairs from the Tatoeba Project, compiled at manythings.org (file dated 2026-02-13) under CC BY 2.0. Per-sentence credits are in `data/fra.txt`. The full paper model is much larger and was trained on WMT; this one is a CPU-sized copy of the architecture.